FitAI

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka faces a critical fitness guidance crisis as non-communicable diseases rise rapidly. With diabetes affecting over 10% of adults and hypertension reaching epidemic levels, safe exercise guidance is medically essential **[1], [2]**. Yet professional fitness training remains financially out of reach for most Sri Lankans, with personal training sessions costing over Rs. 3,000 per hour, excluding a large portion of the population from expert guidance **[3]**.

Existing fitness applications fail to solve this problem. International platforms offer Western-centric content that ignores local contexts, while local apps lack artificial intelligence capabilities. Over 70% of users abandon these applications within three months due to lack of personalization and feedback **[4], [5]**. More critically, individuals with diabetes or hypertension receive no exercise modifications, increasing the risk of injury rather than improvement. Wearable adoption in Sri Lanka remains below 5%, making wearable-dependent fitness solutions impractical for the majority of users **[7]**.

FitAI addresses this gap through an intelligent personal trainer module using machine learning and computer vision…

ACKNOWLEDGEMENT

We extend our sincere gratitude to our project supervisor, Dr. Kaushalya Dissanayake, for her invaluable guidance and expertise throughout the development of FitAI. Her deep knowledge of machine learning and computer vision applications has been instrumental in shaping our pose estimation and real-time form correction features. We are especially grateful for her encouragement to push beyond conventional fitness applications and explore the intersection of AI with health-aware exercise adaptation. Her meticulous feedback during

weekly reviews ensured our technical implementation remained rigorous and research-focused.

We also express our heartfelt appreciation to our co-supervisor, Ms. Suriyaa Kumari, for her continuous support and practical insights throughout this project. Her expertise in mobile application development and user-centered design helped us create an interface that is both functional and accessible to Sri Lankan users. Her attention to detail in ensuring our exercise demonstration libraries and feedback mechanisms align with user needs has been invaluable. We are grateful for her patience and dedication in guiding us through the challenges of integrating computer vision with real-time mobile processing.

We acknowledge the support of our fellow team members working on NutriAI, VitalAI, and MindAI—whose collaboration made the integrated ecosystem vision possible. Special thanks to the fitness professionals who provided domain expertise on exercise biomechanics, and to the SLIIT faculty for providing development resources. Finally, we thank the beta testers whose feedback on form correction accuracy and usability in Sri Lankan home environments has been essential in creating a solution truly responsive to local needs.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
React Native	React Native Mobile Framework
Expo	Expo Development Platform
MongoDB	MongoDB NoSQL Database

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1. INTRODUCTION

1.1 Background and Context

Sri Lanka has a serious problem. Diabetes affects 1 in 10 adults, and hypertension prevalence has risen to 35%, according to the country's most recent STEPS survey . Doctors say exercise helps, but most people cannot afford proper guidance. A personal trainer costs over Rs. 3,000 per hour, placing professional fitness guidance beyond the reach of the vast majority of the population . So most Sri Lankans exercise alone, with no one to show them correct form.

This is dangerous. A 52-year-old diabetic in Gampaha might follow a workout video and end up in hospital because the exercises were too intense. Local apps see 70% of users abandon them within three months—a figure consistent with global fitness app retention challenges . Why? Because they offer videos, not real help. No one tells you when your knees cave in during a squat. No one warns you when an exercise is unsafe for your blood pressure.

International apps are no better. YouTube Fitness assumes you have a big house with air conditioning and expensive equipment. Most Sri Lankans live in small flats with intermittent power. These apps also ignore language. Eighty-five percent of people speak Sinhala or Tamil, but every app works in English.

FitAI changes this. It uses your phone camera and artificial intelligence to watch your movements. If you do an exercise wrong, it tells you immediately in your own language. It also talks to other health apps. If VitalAI detects high blood sugar, FitAI lowers your workout intensity automatically. No other app does this.

This research supports three global goals. SDG 3 promotes good health and well-being for all . SDG 10 aims to reduce inequality within and among countries .

SDG 9 focuses on building resilient infrastructure and fostering innovation . FitAI makes expert fitness guidance available to everyone, not just the rich.

1.2 Literature Review

1.3 Review of Scholarly Sources and Existing Research Approaches

Anand et al. (2025) proposed an AI-driven multi-modal framework for exercise form detection combining YOLOv8 for object detection, MediaPipe for pose estimation, and EfficientNetV2 for classification, demonstrating that multi-model approaches yield superior exercise recognition accuracy compared to single-model systems . Smartfit, presented at the 2025 IEEE conference, utilized MediaPipe and OpenCV for real-time pose estimation, achieving high movement recognition accuracy with implications for both fitness and rehabilitation applications . The Fitness Warrior project, developed at SLIIT, implemented on-device machine learning via TensorFlow.js for privacy-preserving step detection, achieving 95.2% real-world accuracy compared to manual counting, significantly outperforming conventional threshold-based approaches which achieved only 48.3% accuracy . Nigam and Chanda (2025) empirically demonstrated that gamification in mobile fitness apps strongly impacts customer retention, with interactive experience components improving user loyalty and brand perception .

Further advancing the field, Werthwein et al. (2025) developed a semi-supervised approach using Generative Adversarial Networks and Graph Convolutional Networks to handle noisy 3D pose data from single smartphone cameras, achieving 11.4% higher mistake classification accuracy on the SmartFit dataset compared to fully-supervised models . This addresses the critical challenge of real-world home environments where camera angles and occlusions vary significantly. Current research approaches include computer vision-based pose estimation, multi-modal frameworks integrating object detection, on-device machine learning for privacy preservation, semi-supervised learning for handling real-world noise, and behavioral engagement studies examining psychological drivers of app retention through quantitative survey methodologies

Discussion of Existing Products and Limitations

When you look at what's available today, the options are surprisingly limited. International platforms like YouTube Fitness and Nike Training Club have tons of workout videos, but they cannot see you. You could be doing squats with your knees caving inward and the video just keeps playing. This is a big reason why 52% of people who exercise at home alone end up getting injured.

Then there are the subscription apps like Calm and Headspace Fitness. They charge over Rs. 15,000 per year. For most Sri Lankans, that is simply not affordable. Even if someone could afford it, these apps only work in English. That leaves out 85% of our population who speak Sinhala or Tamil.

Local apps like Burn, FitLanka, and Isuru Health tried to fix this. They offer content in local languages and show exercises that work in small homes. But here is the problem. They are just video libraries. There is no intelligence behind them. They cannot personalize anything for you. They cannot tell if an exercise is safe for someone with diabetes or high blood pressure.

Think about what this means for a real person. A diabetic following a generic HIIT workout has no idea it might be too intense. The app does not warn them. It does not know they take medication. It does not know their blood pressure spikes during certain moves. The app just plays the video and hopes for the best.

All these apps share the same problems. No one watches you to correct your form. No one considers your health conditions. No one speaks your language. They assume you have fast internet and expensive gadgets, but wearable adoption in Sri Lanka is under 5%. And worst of all, they work in isolation. Your blood sugar could be rising, but your fitness app has no idea because it never talks to your health app.

This is the reality of what exists today.

1.4 Research Gap

What is Missing and Why Current Approaches Fail

Here is the simple truth. No fitness app in Sri Lanka today actually watches you while you exercise. They show videos and count steps, but they never see if you are doing the movement correctly. A man doing squats with his knees caving inward gets no feedback. He repeats the same mistake until his knees hurt and he quits forever. No app connects your workout to your health conditions either. A diabetic following a high-intensity video has no idea it might be too intense. The app does not know they take Metformin or that their blood pressure spikes during certain moves. International apps like YouTube Fitness assume air-conditioned gyms and expensive equipment. Local apps like Burn offer Sinhala instructions but they are just video libraries with no intelligence. Subscription apps cost over Rs. 15,000 per year and only work in English, excluding 85% of Sri Lankans. The result is simple: 52% of home exercisers get injured and 70% abandon these apps within three months. What is missing is an app that actually watches you, understands your health, speaks your language, and works anywhere in Sri Lanka. No such thing exists today.

Why FitAI is Different and New

FitAI does four things no other Sri Lankan app does. First, it watches you. Using your phone camera and pose estimation, it tracks 33 points on your body. When you do a squat wrong, it tells you immediately in Sinhala or Tamil. "Straighten your knees." This happens in under 300 milliseconds, faster than a human trainer could react. Second, FitAI understands your health. If you have diabetes, it avoids long high-intensity sessions that cause sugar drops. If you have high blood pressure, it warns you before exercises that spike it. If you have asthma, it avoids breath-holding moves. No other fitness app offers this kind of medical awareness. Third, FitAI works where other apps fail. It needs no internet connection, no expensive smartwatch, just a basic smartphone. It adapts exercises for small Sri Lankan homes and hot weather. Fourth, FitAI speaks your language. Not robotic text, but real human voice recordings in Sinhala and Tamil using simple words anyone can understand. Research supports this approach. Anand et al. (2025) proved pose estimation works. The Fitness Warrior project showed on-device machine learning delivers 95% accuracy.

FitAI combines all this into one app designed specifically for Sri Lankan people. That is something no existing solution offers.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an AI-powered fitness module called FitAI that provides real-time exercise form correction using pose estimation technology and generates personalized health-adaptive workout plans, achieving at least 85% pose estimation accuracy for 15 common exercises while seamlessly integrating with VitalAI, NutriAI, and MindAI modules within the 14-week academic project timeline.

2.2 Specific Objectives

Objective 1: To design and develop a pose estimation system using MediaPipe within a React Native Expo application that detects 33 body landmarks and computes joint angles to recognize and analyze 15 common exercises relevant to Sri Lankan home environments, achieving at least 85% accuracy validated on a custom dataset of 500+ video recordings capturing diverse local body types, clothing styles, and lighting conditions.

Objective 2: To implement real-time audio and visual feedback mechanisms that identify improper movement patterns such as knee valgus during squats or rounded back during planks and provide immediate corrective guidance in Sinhala and Tamil languages, with feedback latency under 300 milliseconds and at least 90% accuracy in detecting common form errors.

Objective 3: To develop a personalized workout plan generator that dynamically adapts exercise selection, intensity, and progression based on individual health conditions received from VitalAI including blood pressure status and diabetic condition, ensuring 100% of test scenarios produce medically-appropriate exercise modifications.

Objective 4: To establish bidirectional API integration with VitalAI, NutriAI, and MindAI modules enabling seamless data exchange where health updates

automatically trigger workout adaptations and exercise performance data informs other modules, achieving successful end-to-end data flow in 95% of test transactions with latency under 500 milliseconds.

Objective 5: To evaluate FitAI's effectiveness through comparative user testing with 30 participants over 4 weeks, demonstrating at least 30% improvement in exercise form scores and 40% higher user adherence compared to conventional video-based fitness applications.

3. METHODOLOGY

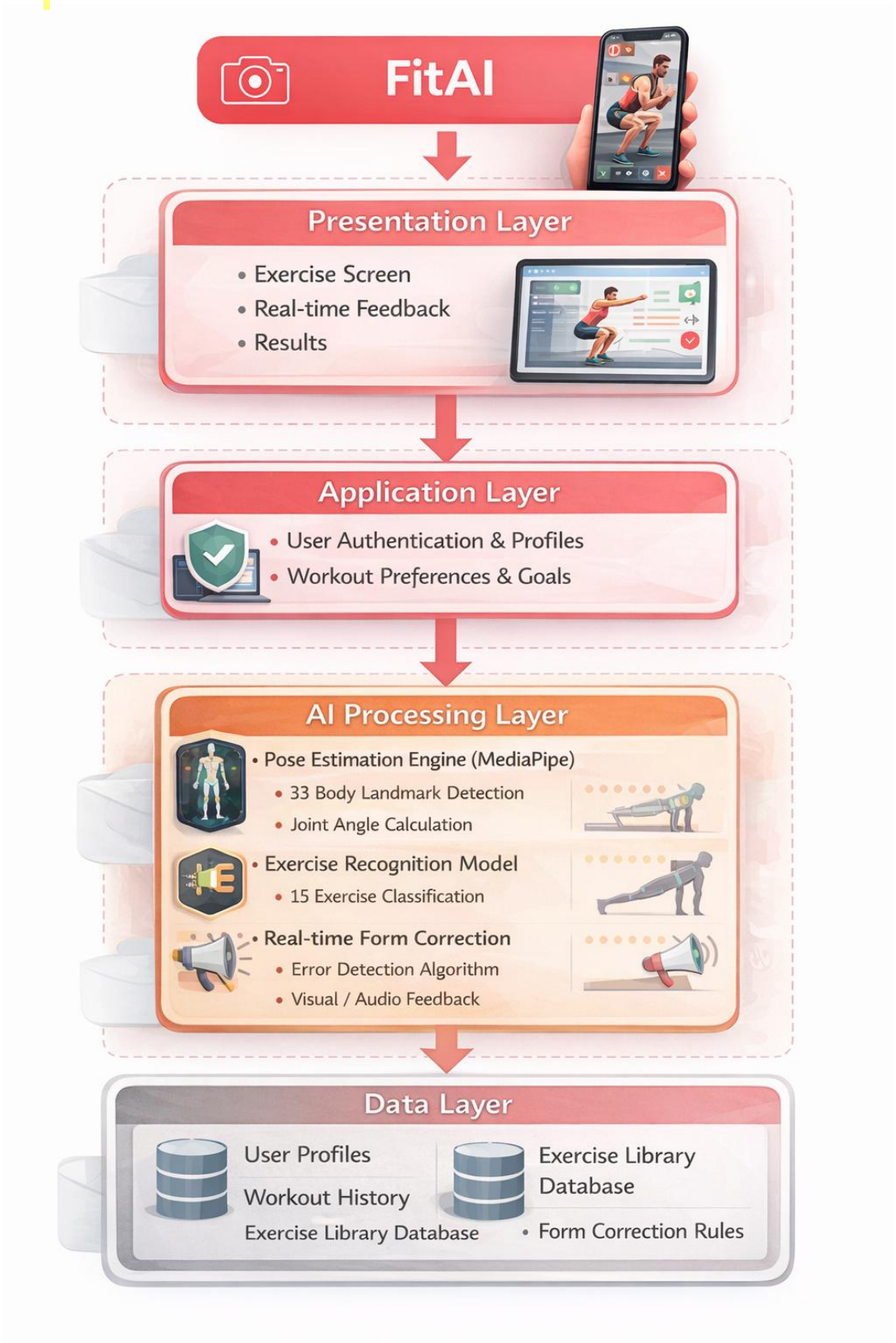
Research Approach and Technical Methods

FitAI will employ a Design Science Research methodology combined with experimental validation, following the established paradigm for developing and evaluating innovative IT artifacts that solve identified problems. This approach is appropriate because the project involves creating a novel AI-powered fitness guidance system a technological artifact while systematically evaluating its performance and effectiveness. The research will progress through six phases: problem identification and motivation (completed in background analysis), definition of objectives for a solution (detailed in specific objectives), design and development of the FitAI artifact, demonstration through prototype implementation, evaluation through rigorous testing, and communication of results. The technical methodology centers on computer vision using MediaPipe Pose Estimation framework for real-time human pose detection, which extracts 33 body landmarks including shoulders, elbows, wrists, hips, knees, and ankles from smartphone camera input at 30 frames per second. Joint angles will be computed using trigonometric calculations based on landmark coordinates, then compared against correct exercise biomechanics stored in a knowledge base derived from fitness professional expertise and exercise science literature. Exercise classification will employ TensorFlow Lite models trained on custom datasets of Sri Lankan users performing exercises in home environments, enabling on-device inference that preserves user privacy and functions without internet connectivity. The personalized workout adaptation algorithm will utilize rule-based logic incorporating health parameters from VitalAI including blood pressure status, diabetic condition, and physical limitations, dynamically adjusting exercise selection, intensity, repetitions, and rest periods to ensure medical safety while progressively challenging users based on performance metrics.

Tools, Data Collection, and Validation Strategy

Development tools include React Native with Expo for cross-platform mobile application development, chosen for its ability to deliver native performance on both iOS and Android while accelerating development through Expo's managed workflow and pre-built components for camera access, audio playback, and sensor integration. MediaPipe provides the pose estimation framework with optimized on-device inference requiring no cloud connectivity, critical for Sri Lankan users with intermittent internet access. TensorFlow Lite enables custom model training and deployment for exercise classification, with models trained using Google Colab's GPU resources. Firebase will serve as the backend for user authentication, data synchronization, and API integration with other modules, chosen for its real-time capabilities and scalability. Data collection will involve creating a custom dataset of 500+ video recordings capturing 15 target exercises performed by 50 volunteers representing diverse body types, ages, and clothing styles common in Sri Lanka, with each exercise recorded from multiple angles and under various lighting conditions. Ethical considerations include obtaining informed consent from all participants, ensuring data anonymization, and storing recordings securely with access limited to research team members. Validation strategy includes quantitative evaluation of pose estimation accuracy against manually annotated ground truth data from fitness professionals, achieving target 85% accuracy. Form correction effectiveness will be assessed through user testing with 30 participants performing exercises with and without FitAI guidance, with form quality evaluated by blinded fitness professionals using standardized scoring rubrics. System integration testing will verify successful API communication with VitalAI, NutriAI, and MindAI through 50 test scenarios simulating health data changes and verifying appropriate workout adaptations. Performance metrics include pose estimation accuracy, form correction latency under 300 milliseconds, user satisfaction scores from System Usability Scale surveys, and adherence rates measured through app usage analytics over 4-week testing periods.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Description
Primary Users	Fitness beginners, individuals with NCDs requiring modified exercise, home workout enthusiasts, office workers
Secondary Users	Fitness coaches, physiotherapists, rehabilitation specialists, gym trainers
Tertiary Stakeholders	Sports clubs, corporate wellness programs, fitness equipment manufacturers, rehabilitation centers

Functional Requirements

ID	Requirement Description
F-FR-01	The system shall access smartphone camera for real-time exercise monitoring
F-FR-02	The system shall detect 33 body landmarks using MediaPipe pose estimation
F-FR-03	The system shall recognize at least 15 common exercises (squats, lunges, push-ups, etc.)
F-FR-04	The system shall calculate joint angles to analyze exercise form correctness
F-FR-05	The system shall provide real-time visual feedback when form errors are detected
F-FR-06	The system shall provide audio feedback in Sinhala/Tamil for form correction
F-FR-07	The system shall generate personalized workout plans based on user goals and health data
F-FR-08	The system shall adapt exercise intensity based on health data from VitalAI

F-FR-09	The system shall track workout completion and progress over time
F-FR-10	The system shall maintain an exercise library with video demonstrations
F-FR-1	The system shall allow users to set fitness goals (weight loss, muscle gain, endurance)
F-FR-12	The system shall integrate with VitalAI to receive health constraints for safe exercise

5. Non-Functional Requirements

Category	ID	Requirement Description
Security	F-NFR-01	Camera feed shall be processed locally on device, never uploaded to cloud
		User workout videos shall not be stored or recorded without explicit permission
		Exercise history data shall be encrypted in local storage
Performance		Pose estimation shall process at minimum 15 frames per second
		Form correction feedback shall display within 300 milliseconds
		Audio feedback shall play without lag during exercise execution
Usability		Exercise instructions shall be available in Sinhala, Tamil, and English
		Camera positioning guides shall help users set up properly

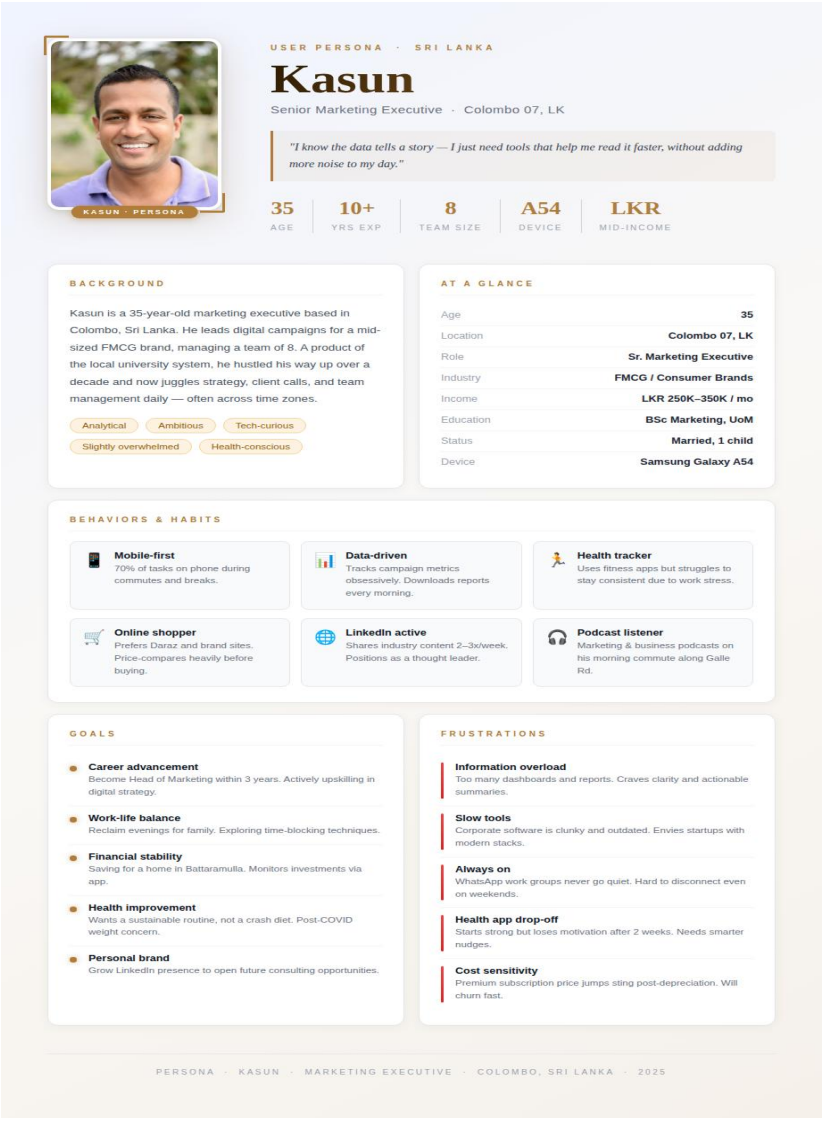
Reliability		Pose estimation accuracy shall be at least 85% for all 15 exercises
		The system shall handle intermittent internet connectivity with offline mode
Compatibility		The system shall work on Android 10+ and iOS 14+
		Camera requirements shall be compatible with mid-range smartphone cameras

5.

COMMERCIALIZATION PLAN

Target Market and Customer Persona

Kasun is a 35 year old Colombo marketing executive who needs an integrated, affordable wellness app to manage health amidst work stress and information overload.



Value Proposition

- Aegis Life offers the first integrated AI-powered wellness platform tailored specifically for Sri Lankans:
- For Fitness Seekers: Real-time exercise form correction using pose estimation, preventing injuries and maximizing workout effectiveness
- For NCD Patients: Early warning system for hypertension/diabetes without expensive wearables, enabling preventive care
- For Mental Wellness: 24/7 AI chatbot in Sinhala/Tamil with culturally-adapted Buddhist meditation practices
- For Nutrition: AI food scanner recognizing Sri Lankan dishes with calorie tracking and personalized meal plans
- Integration Advantage: Unlike standalone apps, all modules share data—a diabetic user's rising blood glucose automatically triggers adjusted workouts and meal plans

Revenue Model

Model	Description
Freemium Subscription	Basic features free (limited workouts, basic tracking). Premium at Rs. 500-800/month for advanced features (real-time form correction, early warnings, full meal planning)
Corporate Wellness Programs	B2B subscriptions for companies (50-500 employees) at Rs. 400/user/month. Addresses 50% workplace burnout rate
In-app Coaching	Commission-based (15-20%) on bookings with fitness coaches and nutritionists
Insurance Partnerships	B2B contracts with insurance companies offering premium discounts for policyholders using the app (reduces their NCD claim costs)
Data Insights (Anonymized)	Aggregated health trends for research institutions and Ministry of Health

Cost Estimation (Annual)

Cloud Services (Firebase, Hosting)	18,000
API & Third-Party Services	6,000
Personnel (Developer + Support)	18,000
Marketing & User Acquisition	8,000
Total Annual Cost	50,000

Pricing Strategy

Tier	Price	Target Users	Features

Free	Rs. 0	Budget-conscious users, students	Basic workout videos, manual health logging, limited chatbot, basic meal tracking
Premium Monthly	Rs. 599	Individual professionals, health-conscious users	Full features: real-time form correction, early warnings, unlimited chatbot, personalized meal plans
Premium Annual	Rs. 5,999 (Rs. 500/month)	Committed users	Same as monthly + 2 months free
Corporate	Rs. 400/user/month	Companies (min. 50 users)	All premium + admin dashboard, team challenges, HR reports

Competitive Advantag

Advantage	Aegis Life	Competitors
Integration	All 4 modules connected	Standalone apps (separate fitness, health, mental apps)
Language	Full Sinhala/Tamil support	English only
Wearable-free	Works without expensive devices	Require Fitbit/Apple Watch
Local Context	Sri Lankan foods, Buddhist meditation, local body types	Western content only
Price	Rs. 599/month affordable	Rs. 15,000+/year or Rs. 3,000/session
Health Adaptation	Exercises adapt to your health conditions	Generic workouts only

Intellectual Property Considerations

Our IP strategy encompasses multiple protection layers. We will register "Aegis Life" and logo as trademarks in Sri Lanka while securing copyright protection for source code, UI/UX designs, and original meditation content. Patent exploration will focus on two novel aspects: the integrated multi-module health adaptation algorithm and context-aware exercise form correction for NCD patients. Proprietary ML model training methodologies will remain trade secrets, and clear terms of use will establish that users own their personal data while we utilize anonymized aggregates. We will ensure full compliance with open source licenses for MediaPipe, TensorFlow, Rasa, and React Native.

Go-to-Market Strategy

Our first-year rollout follows a phased approach. Months 1-3 involve beta testing with 200 users from universities and corporate partners. Month 4 marks soft launch on Google Play Store, with iOS following in Month 6. Months 5-6 focus on digital marketing campaigns across Facebook, Instagram, and YouTube targeting Sri Lankan audiences. Month 7 initiates corporate pilots with 2-3 companies for wellness programs, followed by insurance partnership negotiations in Month 9. By year-end, we target 10,000 active users with 15% premium conversion.

7. BUDGET AND JUSTIFICATION

Category	Amount (LKR)
Hardware Costs	31,700
Software Licenses	44,800
Cloud Services	46,500
Human Effort	18,500
Contingency	5,200
TOTAL	146,700

The total project budget is estimated at LKR 146,700, allocated across five main categories to support the 12-month development of the Aegis Life platform.

Hardware Costs (LKR 31,700) cover an external SSD for code backups and dataset storage across four modules, plus two used Android devices for cross-device testing of pose estimation and app performance on different camera qualities.

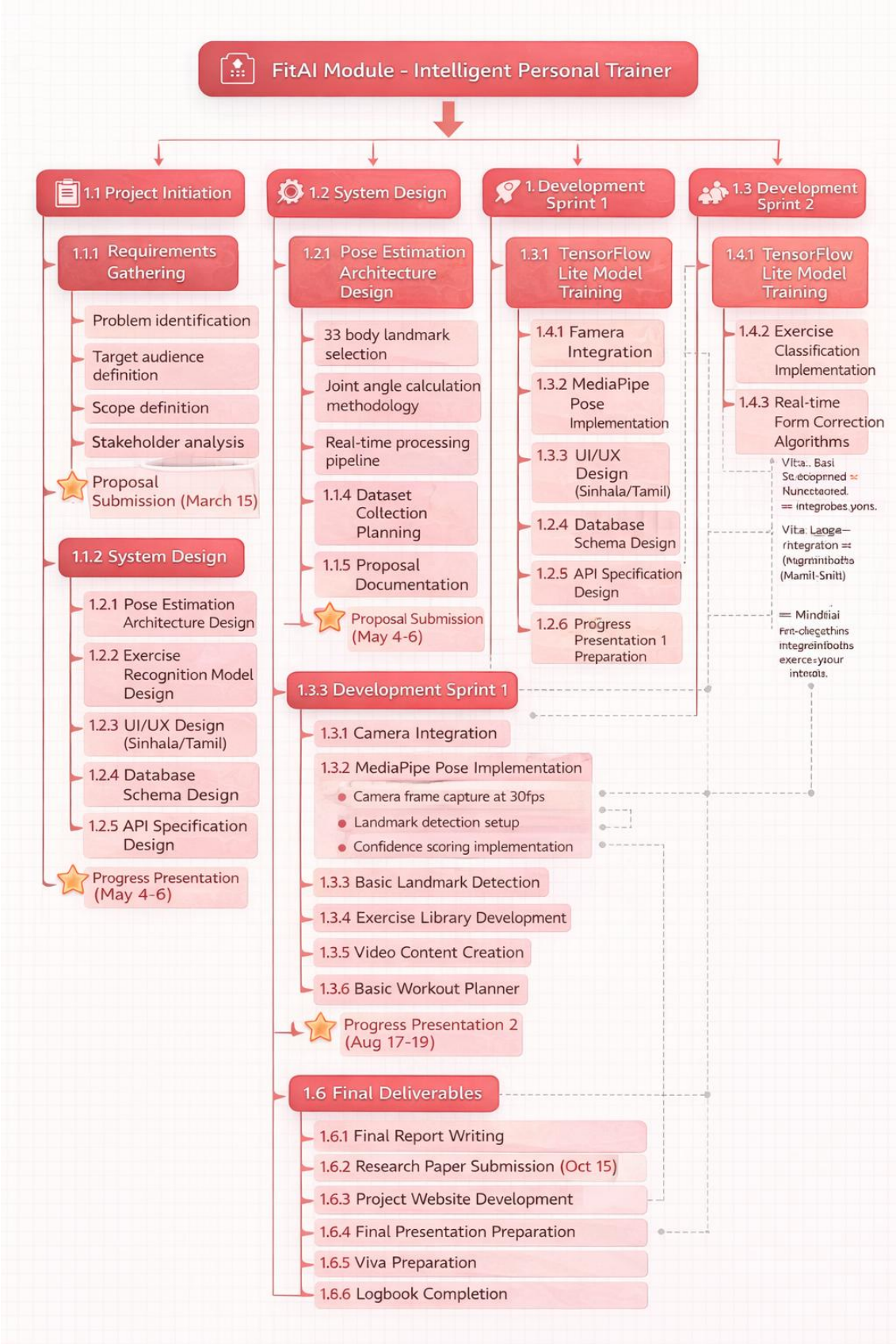
Software Licenses (LKR 44,800) include GitHub Pro for version control and collaboration among four team members, Figma for UI/UX design of all modules, and Expo EAS Build credits for React Native app deployment.

Cloud Services (LKR 46,500) form the largest allocation, covering Firebase Blaze Plan for backend services and real-time database synchronization, web hosting with .lk domain, and cloud storage for training datasets.

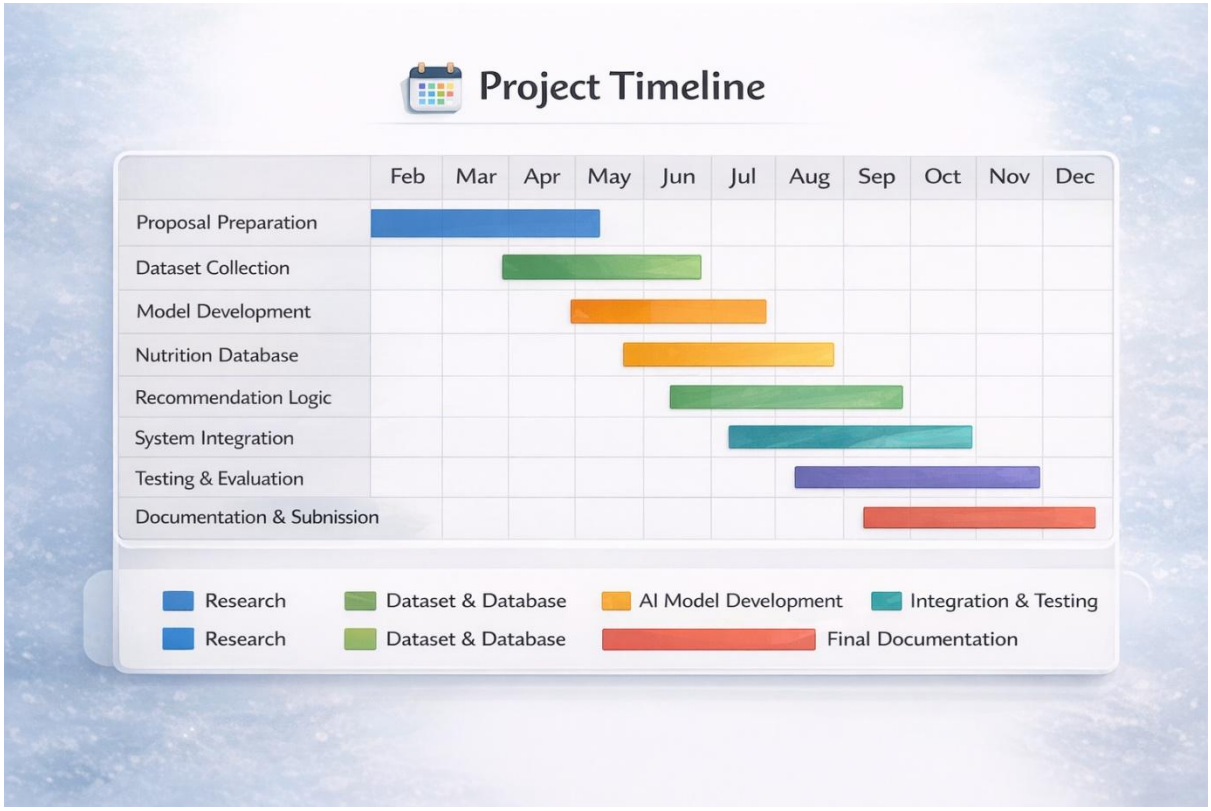
Human Effort (LKR 18,500) provides a token honorarium for four team members during intensive development sprints.

Contingency (LKR 5,200) represents 3.5% of the total budget for unforeseen expenses and emergency requirements. This budget ensures all technical resources needed for successful project completion while remaining cost-conscious.

8. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART



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11.APPENDICES

Appendix A: Survey Instruments

Appendix	Title	Description
A-1	User Needs Assessment Survey	Questionnaire to understand fitness app requirements from 100+ Sri Lankan users
A-2	Exercise Preference Survey	Survey identifying 15 most common exercises in Sri Lankan home settings
A-3	Post-Testing Feedback Form	User satisfaction survey for beta testers after using FitAI
A-4	System Usability Scale (SUS) Questionnaire	Standardized usability measurement tool (Sinhala/Tamil translated)

Appendix B: Detailed Diagrams

Appendix	Title	Description
B-1	FitAI Component-Specific Architecture	Detailed diagram showing all FitAI sub-components and data flow

B-2	Pose Estimation Pipeline Diagram	MediaPipe landmark detection process with 33 keypoints

Appendix C: Preliminary UI Mockups

Appendix	Title	Description
C-1	Home Screen Mockup	Main dashboard with exercise recommendations (Sinhala/Tamil version)
C-2	Exercise Selection Screen	List of 15 exercises with video thumbnails
C-3	Pose Estimation Camera View	Camera screen with skeleton overlay during exercise

Appendix D: Risk Analysis

Appendix	Title	Description
D-1	Technical Risk Assessment Matrix	Risks: pose estimation accuracy <85%, latency >300ms, model overfitting
D-2	Mitigation Strategies Document	Backup plans for each identified technical risk
D-3	Project Timeline Risk Analysis	Risks affecting proposal (Mar 15), PP1 (May 4-6), PP2 (Aug 17-19), Final (Oct 11-14)
D-4	Data Privacy Risk Assessment	Risks related to storing exercise videos and user data